PRACTICE ENVELOPE

Pressure-sensor front-end

Group: _____ Date: _____ Start time: _____

Rules and time

- Write ONE MATLAB script from a blank file, saved as viva.m. It must start with clear; clc; and run from top to bottom without errors. MATLAB help/doc is allowed. No old scripts, notes, internet or AI tools.
- Use exactly the variable names shown in bold below: the examiner runs an automatic checker on your workspace.
- Time: 15 min coding as a team · 3 min checker and team demo · 10 min individual questions (about 1.5–2 min each, on the DFT). Every member must be able to explain every step.

Mission

A pressure waveform (x1, wanted) plus a second channel (x2) with mains hum and an out-of-band spur. Practice case: answers may be released.

Quantity	Value
Sampling frequency, record length	fs = 400 Hz, T = 2 s (so N = fs·T = 800 samples)
ADC	10-bit, input range ±2 V
Continuous signal x1(t)	1·sin(2π·8·t)
Continuous signal x2(t)	0.5·sin(2π·50·t) + 0.6·sin(2π·270·t)
Combination	u[n] = x1q[n] + x2q[n] (addition)
System impulse response h[n]	2-point average: h = [0.5 0.5]

Step 1 · Generate the discrete signal (ADC)

Define fs, nbits, Vmax, T and N, and a time vector t for the sampling instants n/fs. Evaluate x1(t) and x2(t) at those instants. Quantise each with the ADC: delta = 2·Vmax / 2^nbits; xq = delta·round(x/delta), saturated to the range [−Vmax, Vmax − delta].

Variables: **fs, nbits, Vmax, T, N, t, delta, x1q, x2q**

Step 2 · Identify the alias

Exactly one component lies above fs/2. Compute in your script the frequency (Hz) at which it appears after sampling.

Variables: **f_alias**

Step 3 · Combine the signals

Form u from x1q and x2q as stated in the table (sample by sample).

Variables: **u**

Step 4 · System by impulse response

Define h as given. The system output for input u is y = conv(u, h) (full convolution). Also obtain the system's frequency response sampled at N points, H = fft(h, N).

Variables: **h, y, H**

Step 5 · Spectral analysis (DFT)

Take the first N samples of y as yN, compute Y = fft(yN) and the frequency axis f in Hz; plot the one-sided amplitude spectrum 2|Y|/N from 0 to fs/2. Implement YOUR OWN DFT (loop or matrix, from the definition) for the first 64 samples of yN and show that it equals fft. List every spectral peak at least 10% of the tallest peak, in Hz, ascending. The wanted component is the LOWEST peak; the others are unwanted.

Variables: **yN, Y, f, Xd, f_peaks, f_wanted, f_unwanted**

Step 6 · Filter out the unwanted frequencies

Design a filter (IIR or FIR, your choice) that keeps the wanted component within 1 dB and attenuates every unwanted peak by at least 30 dB. Apply it to y. Plot the spectrum before and after.

Variables: **yf, filter_type ('IIR' or 'FIR'), order**

Team demo (3 min)

After the checker runs, be ready to explain: (1) the quantisation step and what it means for accuracy; (2) how you found the alias; (3) why you see these peaks (and, for products, which frequencies appear); (4) what the system did to each peak; (5) your filter choice and its effect.

PRACTICE · INDIVIDUAL CARDS (DFT)

Examiner: cut along the grid lines and place the cards face down. Each member draws one card privately; the answer is given alone, others stay silent.

I1 · DFT by hand

Compute the 4-point DFT of x = [1, 2, 0, -1] by hand, using W = e^(-jπ/2) = -j. Write X[0], X[1], X[2], X[3].

Follow-up: Use a symmetry property of real signals to check X[3] from X[1]. What does X[0] represent?

I2 · Bin and frequency

Your record has N = 800 samples at fs = 400 Hz. Which DFT bin k holds your wanted frequency (8 Hz)?

Follow-up: Which bin is its mirror image, and what frequency (in Hz) does that bin correspond to?

I3 · Why an FFT exists

Your script implemented the DFT sum directly. For a record of N = 800 samples, how many complex multiplications does the direct sum need?

Follow-up: How many does an FFT need (about N·log2 N)? What is the ratio, and why can the FFT be so much cheaper?

I4 · Scale and symmetry

A real tone of amplitude 1 sits exactly on a bin in a record of N = 800 samples. What is |X[k]| at that bin, and what scaling turns |X[k]| into amplitude?

Follow-up: Why do you only plot half of the spectrum for a real signal?

I5 · Resolution and leakage

Your record lasts T = 2 s. What is the bin spacing? Two tones 0.3 Hz apart: are they resolved?

Follow-up: How long must you record to separate them? What happens in |X| if a tone sits at 8 + 0.25 Hz instead of exactly on a bin?

I6 · Convolution and the DFT

Your h has L = 2 taps and u has N = 800 samples. How long is y = conv(u,h)?

Follow-up: If you multiplied the DFTs of u and h to get y, what DFT length must you use? What goes wrong with length N?